A REMARK ON OPENAI'S HOMOGENEOUS DEPTH-FIVE LOWER BOUND FOR ITERATED MATRIX MULTIPLICATION: THE EXPONENT $\sqrt{N}/400$ BECOMES $(1/(4\sqrt{2})-\varepsilon)\sqrt{N}$

[AUTHOR NAME]

Abstract. OpenAI's preprint *Homogeneous depth-five lower bounds for iterated matrix multiplication* proves that every syntactically homogeneous $\Sigma\Pi\Sigma\Pi\Sigma$ circuit computing $\mathrm{IMM}_{n,n}$ over a field of characteristic zero has size at least $n^{\sqrt{n}/400}$ for large n. Re-tuning the parameters of the same proof, and expanding all high-degree factors of an upper product at once instead of one, gives $n^{(1/(4\sqrt{2})-\varepsilon)\sqrt{n}}$, a factor $100/\sqrt{2}\approx 70.7$ in the exponent, with explicit thresholds. Within the framework of the paper's product-rank lemma plus arbitrary expansion, and with $q \asymp n^{-1/2}$, the leading constant $1/(4\sqrt{2})$ cannot be improved (first-order continuum argument). There is no new idea here: the remark is a re-tuning. The IMM-side argument is OpenAI's and is used unchanged. Developed with AI assistance (Anthropic Claude, Opus 5.5) under [AUTHOR NAME]'s direction; independent checks were run by separate AI sessions and are not human peer review.

1. Statement

We write [1] for the preprint, family 135 of `https://github.com/openai/math`, commit `fd4aeeb`. Line numbers refer to `build/sections/*.tex`. Notation ($k, m, \lambda, \rho, v, u, a, b, \alpha, \beta, q, D$, $H_e = \sum_{i=0}^{e} q^{|i-\lambda e|}$, the rank measure R) is that of [1, §§2–3]. Put $L_q = \log(1/q)$.

Theorem 1. *For every $\varepsilon > 0$ there is $n_0(\varepsilon)$ such that for $n \ge n_0(\varepsilon)$ and every field K of characteristic zero, every syntactically homogeneous $\Sigma\Pi\Sigma\Pi\Sigma$ circuit over K computing $\mathrm{IMM}_{n,n}$ has size*

$$S \;\ge\; n^{(1/(4\sqrt{2})-\varepsilon)\sqrt{n}}, \qquad \frac{1}{4\sqrt{2}} = 0.17677\ldots$$

Explicitly, $S \ge n^{c\sqrt{n}}$ holds for

c	1/400	3/320	1/8	3/20	1/6
all $\log n \ge$	17.6	18.24	57.941	112.04	296.735

(natural logarithms; $\log n \ge 57.941$ *means* $n \ge 10^{25.2}$*, and* 296.735 *means* $n \ge 10^{128.9}$*).*

The original theorem states $c = 1/400$ with an unspecified absolute threshold; the first column makes that threshold explicit.

2. The modified proof

M1. The excess s (`02-rank.tex` L57–58). Let s be the largest integer at most $\sqrt{n/2}$ (instead of $\sqrt{n}$) with the parity of n. All other definitions in (eq:parameters) are unchanged. The statement of Lemma 2.1 (lem:parameters) remains true as written, since $\sqrt{n/2} - 2 \ge \sqrt{n}/2$ for $n \ge 95$; the layer schedule of §4 still has $\ell_i \in \{1, 2\}$. Put $t_1 = n/(2s)$ (instead of $t_0 = n/(4s)$) and $x_e = se/(2n)$, so that $x_e < 1/4$ for $e < t_1$ and $\lambda e = e/2 - x_e$.

M2. The bounds on H_e (`03-circuits.tex` L123–205). Replace Lemma 3.3 (lem:degree-estimates) by the following.

Lemma 2. *$H_1 \le 2q^{\lambda}$. For $1 \le e < t_1$ and $x = x_e$,*

$$H_e \le q^x\Big(1 + \tfrac{q^{1-2x}+q}{1-q}\Big)\ (e \text{ even}), \qquad H_e \le q^x\,\tfrac{q^{1/2-2x}+q^{1/2}}{1-q}\ (e \text{ odd}).$$

Date: October 9, 2026.

For every list $e_1,\dots,e_r$ with $\sum e_j \le n$, $\sum_{j:\,e_j<t_1}\log(q^{-x_{e_j}}H_{e_j}) \le C_e + C_o$, where, with $f(z)=q^{1-sz/n}/z$,

$$C_e = \frac{n\max\{f(2),f(t_1)\}+nq/2}{1-q} = \sqrt{n}\,(1+o(1)), \qquad C_o = 4s\Big(\sqrt{q}+\frac{q}{1-q}\Big) = O(n^{1/4}).$$

Proof. For even e the nearest integer to λe is $e/2$ (distance x); the others are at distances $1-x+j$ and $1+x+j$, $j\ge 0$. For odd e the two nearest are at $1/2\mp x$, the others one further step each. Summing geometric tails gives both displays. For even e, $\log(1+y)\le y$ and log-convexity of f on $[2,t_1]$ (as in L184–204) give $q^{1-se/n}\le e\max\{f(2),f(t_1)\}$; summing with $\sum e_j\le n$ and at most $n/2$ even factors gives C_e. For odd e with $x\le 1/8$ the correction factor is $\le (q^{1/4}+q^{1/2})/(1-q)\le 1$; odd e with $x>1/8$ have $e>t_1/2$, so there are at most $4s$ of them, each costing at most $\log\frac{1+\sqrt{q}}{1-q}\le\sqrt{q}+\frac{q}{1-q}$. Since $q=n^{-1/2}(1+o(1))$ by Lemma 2.1, $nf(2)+nq/2=\sqrt{n}(1+o(1))$ and $nf(t_1)=O(n^{1/4})$. □

M3. Expanding all high factors (`03-circuits.tex` L250–299). As in L213–249, g is a sum of at most S scalar multiples of upper products $G=\prod_j Q_j$ with $\sum_j e_j=n$, each Q_j a sum of at most S products of e_j linear forms. Let $X=\{j:e_j\ge t_1\}$ and $E_X=\sum_{j\in X}e_j$, so $|X|\le E_X/t_1$. Expand *every* Q_j, $j\in X$, simultaneously (the standard multilinear expansion); G becomes a sum of at most $S^{|X|}$ products of the unexpanded Q_j ($j\notin X$) and E_X linear forms. Lemma 3.2 (lem:product-rank), Lemma 2 and $\sum_{j\notin X}x_{e_j}=s(n-E_X)/(2n)$, $\lambda=(n-s)/(2n)$ give

$$R(G)\le 2D\,S^{|X|}\exp\big(-L_q s/2 - E_X c' + C_e + C_o\big), \qquad c'=\frac{L_q(n-2s)}{2n}-\log 2. \tag{1}$$

This replaces (eq:all-low-product), (eq:one-high-product) and the conclusion of Proposition 3.1.

M4. Completion (`06-completion.tex` L9–29). Proposition 4.1 (prop:imm) gives $R(\mathrm{IMM}_{n,n})\ge De^{-C_I}$. Carrying the paper's chain in §§4–5 with all terms explicit (the loss in (eq:first-trace), $m\log(1+\beta)$, the prefactor and the word-sum bound $Z\le e^{m/b_0}(2+6n^2e^{\sqrt{n}})$ of lem:word-sum) gives

$$C_I = 2\sqrt{n}+\Big(\tfrac{5}{2}+\tfrac{1}{2\sqrt{2}}\Big)\log n + 1.292 + o(1).$$

The term $\log n/(2\sqrt{2})$ comes from $m\beta=\frac{\sqrt{n}}{2}\,n^{1/\sqrt{2n}}(1+o(1))$ with the new s. Comparing with (1) summed over at most S upper products,

$$(1+|X|)\log S \;\ge\; A + E_X c' \;\ge\; A + |X|\,t_1 c', \qquad A=\frac{L_q s}{2}-K, \quad K=C_I+C_e+C_o+\log 2,$$

hence $\log S\ge\min(A,\ t_1c')$. With $L_q=\frac12\log n\,(1+o(1))$, $s\sim\sqrt{n/2}$ and $t_1\sim\sqrt{n/2}$, both terms are $\frac{1}{4\sqrt{2}}\sqrt{n}\log n\,(1-o(1))$, proving the first part of Theorem 1. Corollary 6.1 (characteristic zero) transfers verbatim.

Thresholds. For the table, every quantity in A and t_1c' is replaced by an explicit majorant or minorant that is monotone in n (worst case of $s\in(\sqrt{n/2}-2,\sqrt{n/2}]$), and the resulting lower bound for $\log S/(\sqrt{n}\log n)$ is evaluated in 60-digit arithmetic and bisected. An independent re-implementation with the exact integer parameters of (eq:parameters) and exact $C_I(n)$ gives the same thresholds (57.941 for 1/8, 296.735 for 1/6; at $\log n=296.7$ the bound is $0.166665<1/6$). The side conditions of Lemma 2.1 hold at $\log n=14$ and beyond.

3. Where the factor 70.7 comes from

step	change	factor
$1/400\to 3/320$	slack in the paper's own chain (it proves any $c<3/320$)	3.75
$3/320\to 1/8$	$q\le n^{-2/5}\to q\approx n^{-1/2}$; $\lambda\ge 3/8\to\lambda\approx 1/2$; threshold and s scaling	13.3
$1/8\to 1/(4\sqrt{2})$	expand all $\lvert X\rvert$ high factors at cost $S^{\lvert X\rvert}$ (M3)	$\sqrt{2}$
total		70.7

With one expansion, the bound is $\min(\sigma/4,1/(16\sigma))$ for $s=\sigma\sqrt{n}$, maximal (1/8) at $\sigma=1/2$. Simultaneous expansion gives $\min(\sigma/4,1/(8\sigma))$, maximal at $\sigma=1/\sqrt{2}$. This is the only step not already present in the paper, and it is a standard multilinear expansion.

4. Limit of the framework

Remark 3. Consider the framework "Lemma 3.2 + expansion of an arbitrary set of factor occurrences (cost $S^{|X|}$)", with the parameter family $\lambda = 1/2 - \sigma/(2\sqrt{n})$ and $q \asymp n^{-1/2}$. To first order (continuum limit, factor degrees $y\sqrt{n}$, the adversary choosing the degrees), the best provable constant is

$$\tau^*(\sigma) = \min\Big(\frac{\sigma}{4},\ \frac{\sqrt{\sigma^2+4}-\sigma}{8}\Big), \qquad \max_{\sigma} \tau^* = \tau^*(1/\sqrt{2}) = \frac{1}{4\sqrt{2}}.$$

So within this framework the leading constant cannot exceed $1/(4\sqrt{2})$. This is a first-order continuum argument, not a theorem. At finite n, exact evaluation of the framework value with exact H_e (uniform-degree adversaries, best number of expanded copies) is consistent with it (e.g. 0.1146 at $n = 10^7$, $\sigma = 1/\sqrt{2}$; the corrections are $O(1/\log n)$). Dropping the constraint $\sum_j i_j = k$ in Lemma 3.2 costs only $e^{O(\sqrt{n})}$.

Remark 4. The real obstacle is on the IMM side. A larger saving on the circuit side needs $q = n^{-\gamma}$ with $\gamma > 1/2$, i.e. $\alpha = n^{-\gamma}$. Exact transfer-matrix evaluation of the word sum of §5 then gives a loss of order $n^{\gamma}/(\sqrt{2}\,\gamma \log n)$, which eventually exceeds the circuit-side gain $\frac{2\gamma-1}{4\sqrt{2}}\sqrt{n}\log n$ (at $n = 10^6$, $\gamma = 0.75$: loss 0.096 against gain 0.088, in units $\sqrt{n}\log n$). Thus $q \gtrsim n^{-1/2}$ is forced by the trace method as used in [1]. The IMM-side bound C_I itself is loose (the exact word sum is about $0.11\sqrt{n}$ at $n = 6 \cdot 10^4$, against about $1.6\sqrt{n}$ in the bound). Tightening it lowers the finite thresholds but not the leading constant; we do not use this.

5. Scope

This is a re-tuning; there is no new idea. The IMM-side proof (§§4–5) is OpenAI's and is used unchanged; we rely on its correctness. The official Lean formalisation (`lean/OAI/Algebra/DepthFive`) covers the original 1/400 statement; nothing here is formalised. Scripts and their expected output are in `scripts/`.

References

[1] OpenAI, *Homogeneous depth-five lower bounds for iterated matrix multiplication*, OpenAI Math Release preprint, September 25, 2026, `https://github.com/openai/math/tree/main/preprints/Homogeneous-depth-five-lower-bounds-for-iterated-matrix-multiplication-September-25-2026`.